QP CODE: 25901404

MAHATMA GANDHI UNIVERSITY, KOTTAYAM MGU-UGP (HONOURS) REGULAR EXAMINATION March2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEMBG202 - DAIRY MICROBIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Part A

Very Short Answer Questions

Answer all questions Each question carries 1 marks

1. Name two microorganisms used as starter cultures in dairy microbiology. [K] [CO1]
Give any two Examples: *Lactobacillus bulgaricus* ,*Streptococcus thermophiles*

2. What is shelf life? [K] / [CO2]
Shelf life is the length of time a food product remains safe and suitable for consumption while maintaining its desired quality under specified storage conditions.

3. What is the role of rennet in cheese production? [K] / [CO3]
Rennet is used in cheese production to coagulate (curdle) milk. It helps separate the milk into solid curds and liquid whey, which is an essential step in forming cheese.

4. Mention the concentration of Formalin in composite milk sample as effective antiseptic and preservative. [K] / [CO4]
0.1% (about **1 mL of formalin per liter of milk**).

5. Define standardized milk. [K] / [CO2]

Standardized milk is milk in which the fat and solids-not-fat (SNF) contents have been adjusted to meet the legal or required standards by adding or removing cream or skim milk.

Part B

Short Answer Questions

Answer any 10 questions Each question carries 2 marks

6. Comment on the sources of contamination of milk. [K] / [CO1]

Animal source – bacteria from the udder, skin, dung, and diseases of the animal.

Environment and equipment – contaminated water, air, utensils, milking machines, and handlers during milking, processing, and storage.

7. Write two sources of microbial contamination in dairy plants. [K] / [CO2]

Contaminated equipment and utensils

Workers/handlers and unhygienic environment

8. Name the uses of whipped cream in food preparation. [K] / [CO3]

- Used as a topping for cakes, pastries, desserts, and beverages.
- Used as a filling and decoration in bakery and confectionery products.

9. Define Pasteurized milk. [U] / [CO4]

Pasteurized milk is milk that has been heated to a specific temperature for a specified time and then rapidly cooled to destroy harmful microorganisms without significantly affecting its nutritional quality.

10. Extend MBRT. [U] / [CO4]

MBRT stands for **Methylene Blue Reduction Test**.

11.What are thermoduric bacteria, and what is their significance in pasteurized milk? [K] / [CO1]

Thermoduric bacteria are heat-resistant bacteria that can survive pasteurization temperatures. Their significance in pasteurized milk is that they may reduce milk quality and shelf life, and some can cause spoilage during storage.

12. Mention any two microbial risks associated with milk distribution. [K] / [CO2]

Growth of pathogenic bacteria due to improper temperature control.
Contamination of milk during handling and transportation.

13. Which microorganisms are mainly responsible for souring of milk? [K] / [CO3]

Lactic acid bacteria, mainly *Lactobacillus* and *Streptococcus* species, are responsible for the souring of milk.

14. Mention two objectives of milk drying. [K] / [CO2]

To increase the shelf life of milk by reducing moisture content.

To reduce storage and transportation costs by decreasing bulk and weight.

15. What are kefir grains composed of? [K] / [CO3]

Kefir grains are composed of a mixture of lactic acid bacteria, yeasts, and proteins embedded in a polysaccharide matrix called kefiran.

16. . Comment on HTST. [U] / [CO1]

HTST (High Temperature Short Time) is a method of pasteurization in which milk is heated to **72°C for 15 seconds** and then rapidly cooled to destroy harmful microorganisms while preserving milk quality.

17. Write two objectives of cream separation. [K] / [CO2]

To obtain cream for the production of butter, ice cream, and other dairy products.
To produce skim milk with reduced fat content.

Part C

Short Essay Questions

Answer any 5 questions Each question carries 6 marks

18. Describe the characteristics of Clostridium species associated with milk and dairy products.

[U] / [CO3]

Morphology of Clostridium species, Spore formation, Anaerobic nature.
Occurrence in milk, Spoilage activity
Public health importance

19. Describe the production process of evaporated milk. [K] / [CO2]
Selection and filtration of milk, Standardization
Preheating, Evaporation, Homogenization, Cooling and stabilization
Canning and sealing, Sterilization, Cooling and storage

20. What is homogenisation of milk?What is the difference between single-stage and two-stage homogenization? [U] / [CO1]
Define Homogenisation .Mention any four difference

21. Explain in detail the types of spoilage of milk. [K] / [CO1]
Types of spoilage of milk and characteristics

22. Describe the role of lactic acid bacteria in milk fermentation. [U] / [CO3]

Lactic acid bacteria (LAB) play an important role in the fermentation of milk and production of dairy products such as curd, yogurt, cheese, and kefir. **Functions of lactic acid bacteria:**Examples

23. Explain the characters and preparation process of acidophilus milk. [U] / [CO3]
Acidophilus milk is a fermented milk product prepared by fermenting milk with *Lactobacillus acidophilus*. It is valued for its therapeutic and probiotic properties.
Characteristics, Preparation stages

24. Write in detail about Agar Slice Methods for milk processing equipment. [U] / [CO4]

The agar slice method is a microbiological technique used to examine the sanitary condition and microbial contamination of milk processing equipment and plant surfaces. It helps in evaluating the effectiveness of cleaning and sanitation procedures in dairy plants.

Part D

Essay Questions Answer any 1 questions

Each question carries 15 marks

25. Explain the scope and importance of Dairy Microbiology. [U] / [CO1]

Dairy microbiology deals with the study of microorganisms present in milk and milk products, including useful, spoilage, and disease-causing microbes.

Scope

1. Study of microorganisms in milk and dairy products.
2. Production of fermented products like curd, yogurt, and cheese.
3. Detection and prevention of milk spoilage.
4. Dairy plant sanitation and hygiene.
5. Quality control and microbiological testing of milk.
6. Preservation methods such as pasteurization and refrigeration.
7. Control of milk-borne diseases and pathogens.

Importance

1. Ensures safe and hygienic milk supply.
2. Increases shelf life of milk products.
3. Helps in manufacture of fermented dairy products.
4. Maintains quality, flavor, and nutritive value.
5. Prevents spoilage and economic losses.
6. Protects public health from milk-borne diseases.

26. Explain the characteristic properties of milk with reference to its composition and properties. [K] / [CO1]

Composition of Milk. Characteristic Properties of Milk

1. Physical Properties
2. Chemical Properties
3. Biological Properties